Práctica 1: Diseño de controladores

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

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Información general

Nombre del alumno: Marco Antonio Campoy Alegria

Número de control: 21212145

Correo institucional: L21212145@tectijuana.edu.mx

Asignatura: **Modelado de Sistemas Fisiológicos**

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos de la simulación

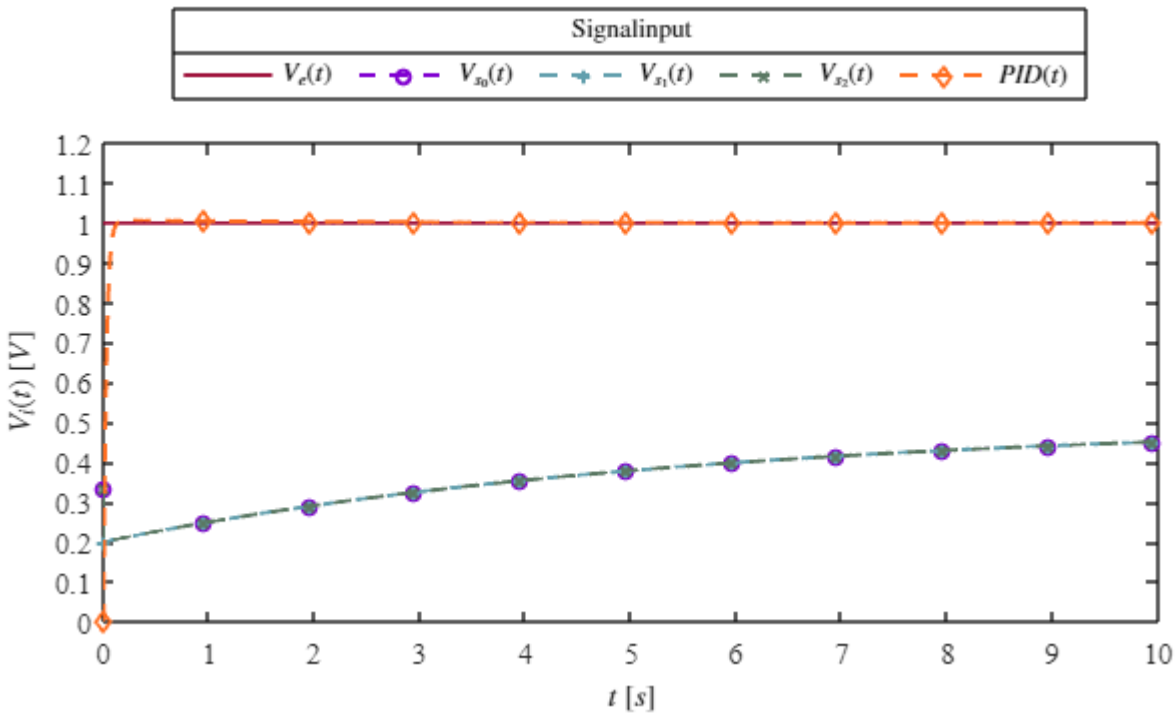
```
clc; clear; close all; warning('off','all')
tend = '10';
file = 'Sistema';
open_system(file);

parameters.stoptime = tend;
parameters.Solver = 'ode15s';
```

```
parameter.Maxstep = '1E-3';
set_param('Sistema/Vs0(t)','VectorFormat','1-D array')
```

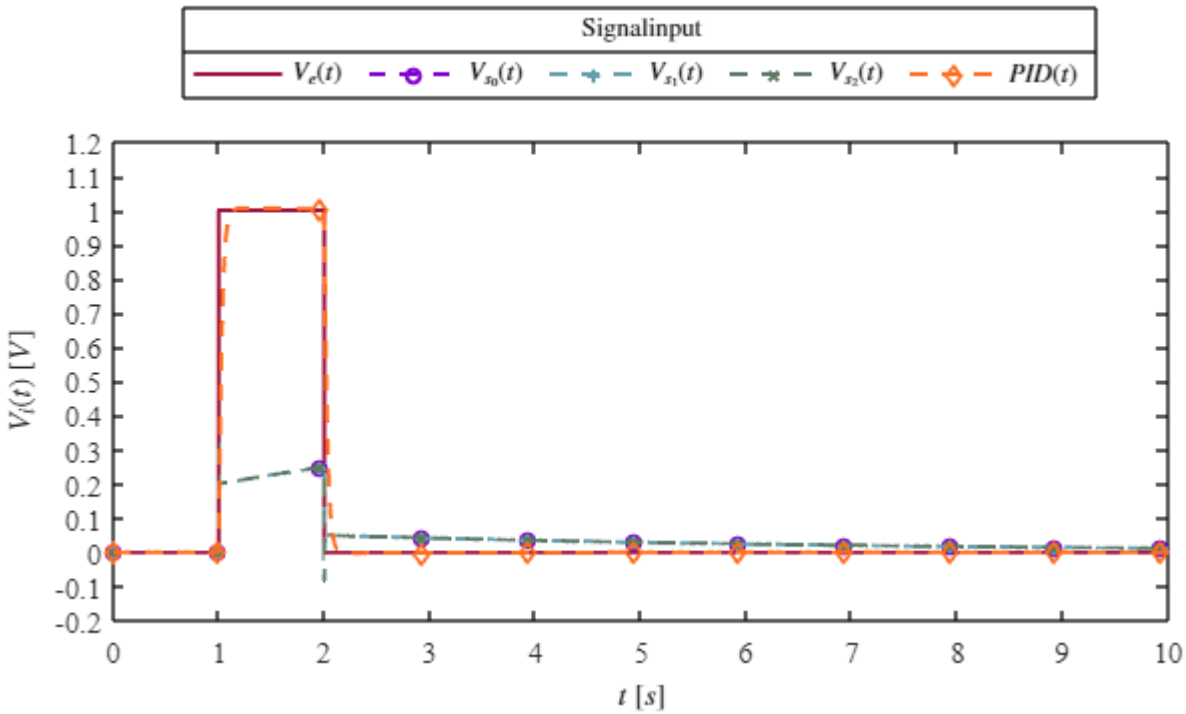
Respuesta al escalón

```
Signal = 'Step';
set_param('Sistema/S1','sw','1');
set_param('Sistema/Ve(t)','sw','1');
x1 = sim(file,parameters);
plotsignals(x1.t,x1.Ve,x1.Vs0,x1.Vs1,x1.Vs2,x1.PID,Signal)
```



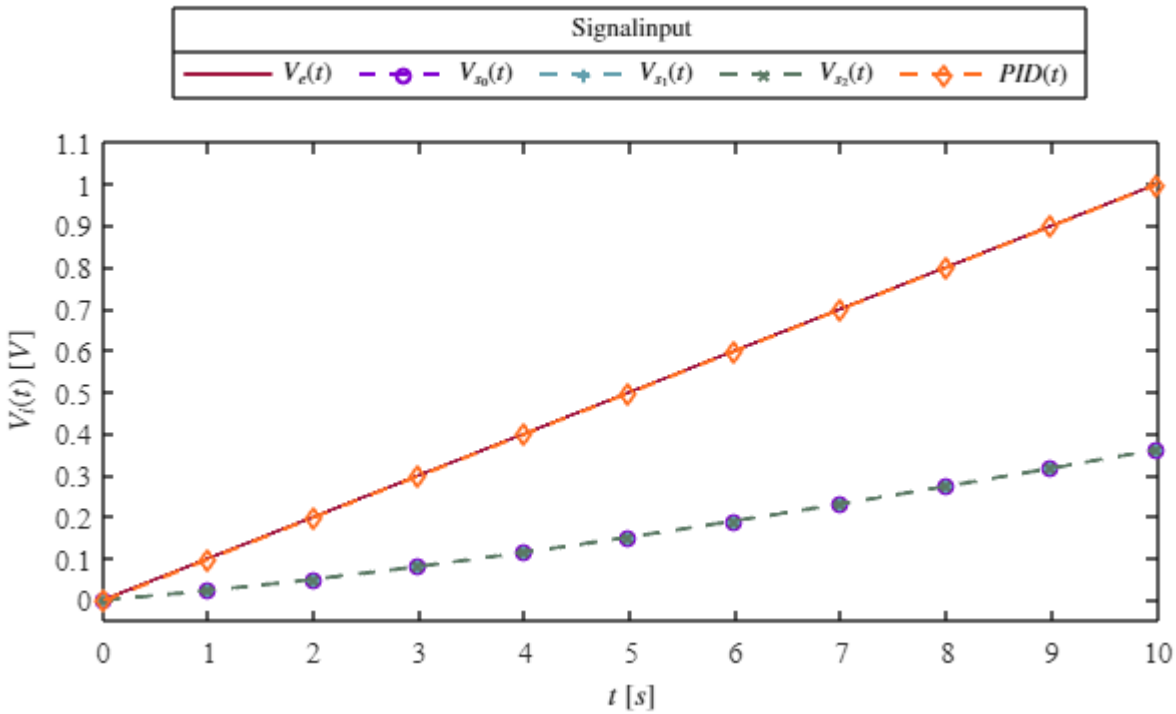
Respuesta al impulso

```
Signal = 'Impulse';
set_param('Sistema/S1','sw','0');
set_param('Sistema/Ve(t)','sw','1');
x2 = sim(file,parameters);
plotsignals(x2.t,x2.Ve,x2.Vs0,x2.Vs1,x2.Vs2,x2.PID,Signal)
```



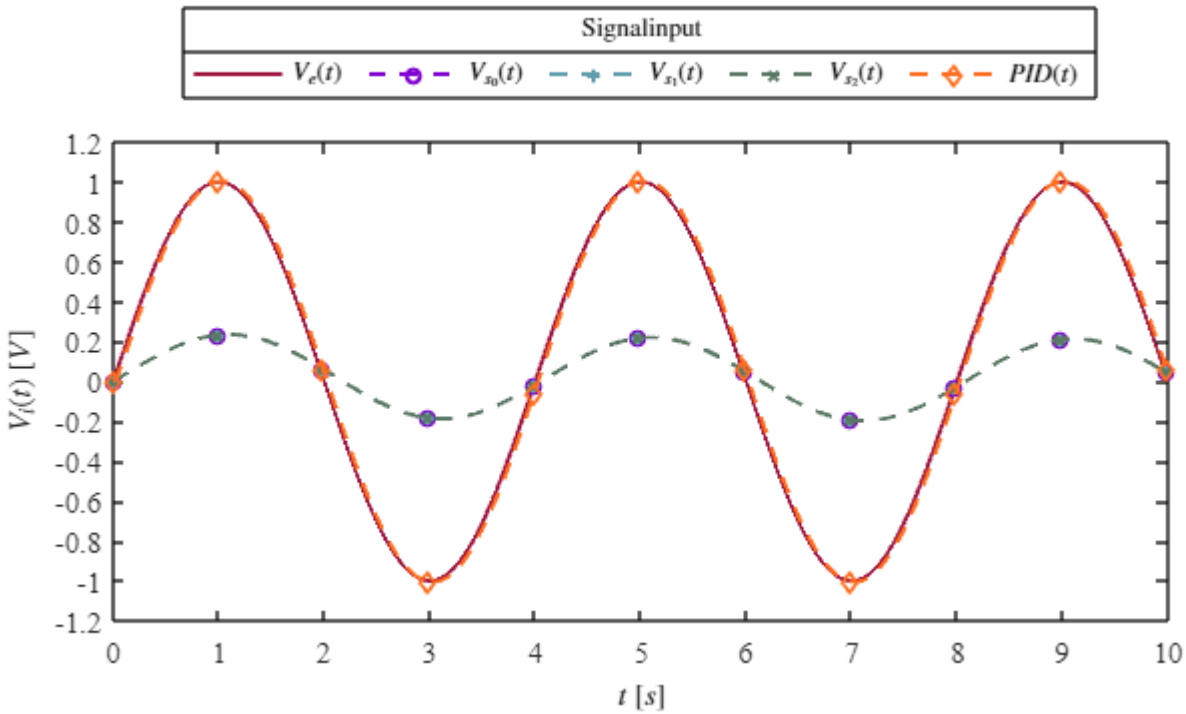
Respuesta a la rampa

```
Signal = 'Ramp';
set_param('Sistema/S2','sw','1');
set_param('Sistema/Ve(t)','sw','0');
x3 = sim(file,parameters);
plotsignals(x3.t,x3.Ve,x3.Vs0,x3.Vs1,x3.Vs2,x3.PID,Signal)
```



Respuesta a la función sinusoidal

```
Signal = 'Sin';
set_param('Sistema/S2','sw','0');
set_param('Sistema/Ve(t)','sw','0');
x4 = sim(file,parameters);
plotsignals(x4.t,x4.Ve,x4.Vs0,x4.Vs1,x4.Vs2,x4.PID,Signal)
```



Funcion: Respuesta a las señales

```
function plotsignals(t, Ve, Vs0, Vs1, Vs2, PID, Signal)
set(figure(),'Color','w')
set(gcf, 'units','centimeters','position',[1,1,18,10])
set(gca,'FontName','Times New Roman','FontSize',11)
hold on, grid off; box on;

Loscolors = [160, 21, 62;
    129, 0, 209;
    98, 159, 173;
    90, 120, 99;
    255, 109, 31]/255;
colororder(Loscolors)

 plot(t,Ve,'-',t,Vs0,'--o',t,Vs1,'--+',t,Vs2,'--x',t,PID,'--d', ...
     'LineWidth',1.5,'MarkerSize',5,'MarkerIndices',1:1000:length(t));

L=legend('$V_{e}(t)$','$V_{s_0}(t)$','$V_{s_1}(t)$','$V_{s_2}(t)$','$PID(t)$');

set(L,'Interpreter','Latex','FontSize',10,'Location','northoutside',...
      'Orientation','horizontal','Box','on')

title(L,['Signal','input'],'FontSize',10)
    xlabel('$t$ $[s]$','Interpreter','Latex','FontSize',11)
    ylabel('$V_i(t)$ $[V]$','Interpreter','Latex','FontSize',11)
    if Signal =="Step"
        xlim([0,10]); xticks(0:1:10)
```

```
      ylim([0,1.2]); yticks(0:0.1:1.2)

elseif Signal == "Impulse"
    xlim([0,10]); xticks(0:1:10)
    ylim([-0.2,1.2]); yticks(-0.2:0.1:1.2)

elseif Signal == "Ramp"
    xlim([0,10]); xticks(0:1:10)
    ylim([-0.05,1.1]); yticks(0:0.1:1.2)

elseif Signal == "Sin"
    xlim([0,10]); xticks(0:1:10)
    ylim([-1.2,1.2]); yticks(-1.2:0.2:1.2)
end

exportgraphics(gcf,[Signal,'.pdf'],'ContentType','vector')
%exportgraphics(gcf,[Signal,'.png'],'ContentType',600);
%print(Signal,'-dsvg', 'r600');
%print(Signal, 'depsc','-r600');
end
```